

The Clock at Zero Gradient

Rest Mass as the Bounded-Continuation Term, and Why Its Value Remains Unexplained

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Every quantity is quoted verbatim from execution. This paper closes one open target of the preceding paper in this series and fails to close another; both outcomes are reported at equal length. A claim made during the investigation — that a uniform massless field is frozen — was found to be imprecise and is corrected in place in Section 4 rather than quietly repaired.

Abstract

The preceding paper in this series constructed a single local update rule on the edges of a graph, from which the heat, wave, and telegrapher equations follow as parameter limits with exact conservation. It left one term unreachable: a rest mass, being a contribution that survives when every gradient has vanished. This paper supplies that term and shows it is not an import but a requirement.

The argument turns on a distinction that must be stated precisely, because the loose form of it is wrong. The governing constraint is not that a state must differ from its successor — a self-map is perfectly representable, and zero change is a value like any other. The forbidden object is a state whose successor relation is undefined: not zero, but the absence of the relation itself. Restated for a field, the question is not whether the field changes when every spatial distinction vanishes, but whether its evolution operator remains defined and admits a bounded persistent solution.

Measured on a perfectly uniform configuration, a massless field has exactly two options and neither is a persistent state. With zero initial rate it is static, holding at 1.0000 indefinitely. With nonzero initial rate it grows linearly without bound — measured climbing from 1.0010 to

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5.0000 over the run and continuing, since the uniform massless solution is a straight line in time. Static or runaway; no bounded nontrivial option exists. A massive field in the identical configuration oscillates and stays bounded for every initial condition, with maximum amplitude 1.0002 and 1.0000 in the two cases tested. The mass term is what makes the gradient-free mode a persistent state.

The rate of that oscillation is not free. At zero wavenumber the dispersion relation gives an angular frequency equal to the mass times the square of light speed divided by the reduced Planck constant — the Compton rate, 7.7634×10^{20} radians per second for an electron. Simulated directly, the measured frequency matches the predicted value to 0.006 percent. Rest mass is the intrinsic rate at which a field with no spatial structure continues to evolve.

One prediction follows that is not built into the argument, and it is checked against the particle table. If mass exists because a gradient-free mode needs an internal clock, then anything that can be brought to rest requires mass and anything that can never be brought to rest does not. Relativity supplies the independent fact that massless particles have no rest frame. The correspondence holds without exception across all nine particles examined. In the edge operator the change is one term, reading the value at an edge's endpoints rather than their difference, and the cost is exact: the shift symmetry under adding a constant to the field everywhere is broken, verified by direct computation. Gaining mass and losing pure relationality are the same event.

The second target is not reached. An attempt to obtain the observed mass ratios as the spectrum of a natural operator failed against every candidate tested: the observed electron-muon ratio is 206.77 while the largest ratio produced by any standard spectrum — box, oscillator, sphere, drum, hydrogen — is 5.27. The pattern is neither polynomial nor geometric nor log-uniform. The Koide relation is examined with statistical controls and clarified: it is a single constraint reducing three masses to two parameters, it predicts the tau to 0.038 percent given the other two, and 1.2×10^{-5} of random triples match it as well. It is a genuine open puzzle and it is not evidence for this framework. The paper therefore establishes the necessity of the mass term and leaves its magnitude exactly where the Standard Model leaves it: unexplained.

1 The Target

The preceding paper in this series built a dynamics on the edges of a graph. A density sits at each vertex, a flow sits on each edge, and one rule updates the flow: it retains a fraction of

itself and responds to the density difference across the edge. Vertices are not dynamical; they enforce that flows balance. From this, conservation follows exactly for all parameters, and the retention coefficient selects between diffusion, propagation, and damped propagation.

The construction had a stated boundary. Every term in it is driven by a difference, so wherever the difference vanishes the driving vanishes. A rest mass is precisely a contribution that does not vanish under those conditions — an intrinsic quantity that persists with nothing to compare against. The paper recorded this as unreachd and identified what would be required: a term that survives at zero gradient, which is a modification rather than a special case.

This paper supplies that term. The claim is not that the framework predicts any particular mass. It is that the term's presence is forced by the same constraint that generated the rest of the construction, and that its absence produces a configuration the constraint excludes.

2 The Constraint, Stated Precisely

The constraint governing this program is usually stated as: every admissible state admits a successor distinct from itself. That statement is serviceable in most contexts and imprecise here, and the imprecision matters enough to correct before the argument depends on it.

Zero is not null. A self-map — a state whose successor is itself — is entirely representable. Zero change is a value inside the algebra, as legitimate as any other value. A vanishing gradient is a legitimate field configuration. Mathematics handles zero without difficulty.

What mathematics cannot process is the absence of the relation. A state for which no successor is defined is not a state with a small change; it is a state where the evolution operator has no output. This is the difference between a variable holding zero and a variable holding nothing at all, and only the second is a failure of the structure rather than a value within it.

The constraint, corrected. *Every admissible state must participate in a defined continuation relation. Further, for a state to persist, that relation must admit a bounded solution.*

The second clause is not decoration. A relation that is defined but whose only solutions grow without limit does not describe anything that can persist: an unbounded configuration carries unbounded energy and is not a physical state. Persistence requires both that the operator be defined and that it admit bounded evolution. Section 4 shows that a massless field, in the gradient-free limit, satisfies the first and fails the second.

Restated for a field. The question is not whether the field changes when all spatial distinctions vanish. It is whether its evolution operator remains defined and admits a bounded persistent solution in that limit.

3 Where Mass Sits

The relativistic scalar field equation contains three kinds of term, and they differ in what they read.

$$\partial^2\phi/\partial t^2 - c^2 \nabla^2\phi + (mc^2/\hbar)^2 \phi = 0$$

The first term is the second time derivative. The second reads the spatial curvature — a difference of differences. The third reads nothing but the field value at the point. It is the only term in the equation that involves no derivative at all.

That distinction is the entire subject of this paper. A derivative term reads a relation: it asks how the field here compares to the field nearby, and it returns zero when the comparison returns nothing. The mass term reads an amplitude: it asks what the field is, not how it differs, and it returns a nonzero result whenever the field itself is nonzero — including when every comparison in the system has gone flat.

4 The Measurement, and a Correction

Take a perfectly uniform field configuration. The spatial curvature is identically zero everywhere, so the second term of the field equation contributes nothing. What remains is the behaviour of the first and third terms alone.

A correction is owed here. An earlier statement of this result, made during the investigation, claimed that a uniform massless field is frozen. That is imprecise. With the curvature term absent, the massless equation reduces to a vanishing second time derivative, whose general solution is linear in time: a constant plus a rate. The field is frozen only when that rate is zero. The correct statement is that the massless uniform mode has two branches and neither is a persistent state.

Configuration	ϕ initial	ϕ final	max $ \phi $	Persistent?
massless, zero rate	1.0000	1.0000	1.0000	static

Configuration	ϕ initial	ϕ final	$\max \phi $	Persistent?
massless, nonzero rate	1.0010	5.0000	5.0000	runaway
massive, zero rate	0.9984	-0.9821	1.0002	bounded
massive, nonzero rate	0.9994	-0.9769	1.0000	bounded

The massless branches. With zero initial rate the field holds at 1.0000 for the entire run. Its successor equals its state at every step. With nonzero initial rate it climbs — measured from 1.0010 to 5.0000 across the run, and analytically it continues without limit, since a straight line in time has no bound. The finite run does not itself demonstrate unboundedness; the closed-form solution does, and the measured linear climb is consistent with it. Neither branch is a persistent physical configuration: one is a state that has stopped, the other carries energy growing without limit.

The massive case. With the mass term present the same configuration oscillates and remains bounded for both initial conditions, with maximum amplitudes 1.0002 and 1.0000. The evolution operator is defined, the solution is bounded, and the state persists while continuing to evolve. This is the only one of the four cases that is a persistent state with ongoing evolution.

The conclusion. *The mass term is what makes a gradient-free field mode a bounded persistent state. Without it the uniform mode is either stopped or divergent, and neither satisfies the requirement that an admissible state persist under a defined continuation.*

5 The Rate Is Fixed, and It Is Measured

The mass term does not merely permit oscillation; it sets its frequency. At zero wavenumber the relativistic dispersion relation reduces to an angular frequency determined entirely by the mass.

$$\omega^2 = c^2 k^2 + (mc^2/\hbar)^2 \rightarrow \text{at } k = 0: \omega = mc^2/\hbar$$

Simulated directly with a mass-squared coefficient of 400, the predicted frequency is 20.0000 and the frequency recovered from zero crossings is 20.0011 — an error of 0.006 percent. The oscillation rate is exactly the mass coefficient, as the dispersion relation requires.

Particle	Compton angular frequency	Period
electron	$7.7634 \times 10^{20} \text{ rad/s}$	$8.0933 \times 10^{-21} \text{ s}$
proton	$1.4255 \times 10^{24} \text{ rad/s}$	$4.4077 \times 10^{-24} \text{ s}$

This frequency is a standard quantity, associated with Zitterbewegung, and it is not introduced here. What the argument adds is an interpretation of what it is: the rate at which a field with no spatial structure continues to evolve. Rest mass is that rate, expressed in mass units. A rest frame is not a frame in which nothing happens; it is the frame in which spatial transport has been removed and only the field's own evolution remains visible.

6 A Prediction, Checked Against the Particle Table

The argument makes a prediction it was not constructed to make, which is the only kind worth reporting. If the mass term exists because a gradient-free mode requires a bounded internal evolution, then it is needed exactly by those things that can be gradient-free — that is, by anything that can be brought to rest. Anything that can never be brought to rest never encounters the problem and requires no such term.

Relativity supplies the independent half of the test. A massless excitation propagates at the invariant speed in every frame and has no rest frame; its worldline has vanishing proper time, so no internal clock is available to it and none is required. A massive excitation has a rest frame by construction.

Particle	Rest mass	Has rest frame	Consistent
photon	0	no	yes
gluon	0	no	yes
graviton (hypothetical)	0	no	yes
electron	$9.109 \times 10^{-31} \text{ kg}$	yes	yes
proton	$1.673 \times 10^{-27} \text{ kg}$	yes	yes
W boson	$1.433 \times 10^{-25} \text{ kg}$	yes	yes

Particle	Rest mass	Has rest frame	Consistent
Z boson	1.626×10^{-25} kg	yes	yes
Higgs	2.23×10^{-25} kg	yes	yes
neutrino	$\sim 10^{-37}$ kg	yes	yes

Nine of nine, no exceptions. The correspondence is exact in both directions: massless if and only if no rest frame, massive if and only if a rest frame exists. This is a known relativistic fact; what the argument contributes is a reason for it. A thing that can hold still must carry its own clock or it has no admissible persistent state. A thing that can never hold still needs no clock, because it is never in the situation that demands one.

7 The Edge Operator, Completed

The preceding paper's operator updates an edge flow from its own retained value and the density difference across it. Where that difference vanishes, the second contribution vanishes and nothing generates flow. The completion is one additional term.

$$J(t+1) = \alpha \cdot J(t) - \beta \cdot (\rho_j - \rho_i) - \mu^2 \cdot (\rho_j + \rho_i) / 2$$

The new term reads the sum of the endpoint values rather than their difference. It therefore does not vanish when the endpoints agree, which is exactly the property required.

The cost is a symmetry, and it is measured. A theory built only from differences cannot detect a constant added to the field everywhere: shifting all values by the same amount leaves every difference unchanged. Adding the value-reading term destroys that invariance. Tested directly by adding a constant K to both endpoints:

K added everywhere	Massless flow	Massive flow
0	0.2000	-0.8000
1	0.2000	-1.2000
5	0.2000	-2.8000
10	0.2000	-4.8000

The massless flow is invariant to four decimal places at every shift. The massive flow moves with the shift. A massless field knows only relationships and has no absolute reference; a massive field knows where it stands. In field-theoretic language the mass term breaks the shift symmetry of the massless theory, which is standard. The point here is the direction of the implication: the term was not added in order to break the symmetry, it was added because a gradient-free mode needs a bounded continuation, and the symmetry breaking is the unavoidable consequence. Acquiring mass and losing pure relationality are one event, not two.

8 The Second Target: Not Reached

Establishing that a mass term must exist says nothing about its magnitude. This section reports an attempt to obtain the magnitudes and its failure, because the failure bounds the problem and because reporting only the successful half would misrepresent the state of the work.

The approach. If mass is a rest-frame frequency, the mass spectrum is a frequency spectrum, and frequencies are quantised when a wave is confined to a bounded domain — they become the eigenvalues of that domain. The natural test is therefore whether the observed mass ratios match the spectrum of any natural operator. Ratios are used rather than values because they are independent of unit conventions.

The result. Every candidate fails, and not narrowly.

Spectrum	Second / first	Third / second	Third / first
observed, charged leptons	206.768	16.817	3477.228
particle in a box	4.000	2.250	9.000
harmonic oscillator	3.000	1.667	5.000
sphere Laplacian	3.000	2.000	6.000
circular drum	5.269	2.458	12.949
hydrogen binding	0.250	0.444	0.111
linear	2.000	1.500	3.000

The observed first ratio is 206.8; the largest produced by any standard spectrum is 5.27. Simple eigenvalue spectra grow polynomially, and the charged leptons span nearly four orders of magnitude across three states. The pattern is also not geometric — the logarithmic spacings are 5.3316 and 2.8224, differing by 47 percent — and it is not uniform in the square-root coordinate either. To reproduce the observed ratios an operator would need eigenvalues growing far faster than polynomially between the first two states and far slower between the next two, a spacing pattern that changes character mid-spectrum. No single simple domain does this.

What the failure means. This is evidence against the direction, not merely absence of evidence for it. The proposal that mass values are eigenvalues of a small closed operator predicts that the ratios should look like a spectrum, and they do not look like any spectrum tested. Stating that mass is an eigenvalue without exhibiting the operator renames the problem rather than solving it, and the renaming is not free: it makes a prediction about the shape of the spectrum, and that prediction fails.

9 The Koide Relation, With Controls

One empirical relation among the lepton masses is precise, unexplained, and has stood since 1981. Because it is the obvious thing to reach for at this point in the argument, it is examined here with the controls the rest of this program applies to its own claims.

The relation states that the sum of the three charged-lepton masses, divided by the square of the sum of their square roots, equals two-thirds.

Quantity	Value
measured ratio	0.6666605115
two-thirds	0.6666666667
relative deviation	9.233×10^{-6}

First control: other sectors. The up-type quarks give 0.848981 and the down-type quarks 0.731428, against two-thirds. The relation is specific to the charged leptons. One hit and two misses is consistent either with the leptons being special or with coincidence.

Second control: is the target value special? The quantity is bounded below by one-third, attained when all three masses are equal, and approaches one when a single mass dominates.

Two-thirds is the exact midpoint of that range, and the distribution of the quantity for random mass triples piles up near the middle: 2.34 percent of random triples land within 0.01 of two-thirds. Landing near two-thirds is not surprising. Landing on it this precisely is the only surprising part.

Third control: how precise is precise? Two million random mass triples spanning the observed range were drawn. Twenty-four matched two-thirds better than the leptons do — a rate of 1.2×10^{-5} , or roughly one in eighty-three thousand.

What the relation actually constrains. Written in the standard parameterisation, in which the square roots of the masses are proportional to one plus root-two times the cosine of an angle advanced by one third of a turn between generations, the relation holds identically for every value of that angle — verified here at several values, each returning two-thirds to twelve decimal places. The relation is therefore one constraint on three masses, leaving two free parameters. It is not a spectrum. Its genuine content is predictive: fitting the angle to reproduce the electron-muon ratio and then computing the third mass gives 1777.535 MeV against an observed 1776.860 MeV, an error of 0.038 percent.

Verdict. A genuine open puzzle, and not evidence for this framework. The framework predicts that a mass term must exist; it does not predict this relation, and this relation does not confirm it. The one-in-eighty-three-thousand figure is not a significance, because the functional form was selected after seeing the masses and the space of simple functions of three numbers is large. Recorded as an open observation.

10 What This Establishes

The boundary is now sharp, and it is worth stating in both directions.

Established. A field admitting gradient-free configurations requires a non-derivative term in its evolution equation, because without one the uniform mode is either static or unbounded and neither persists. That term is the mass term. Its coefficient sets the rate at which the gradient-free mode evolves, which is the Compton frequency, verified to 0.006 percent. The requirement predicts that mass is needed exactly by things that can be brought to rest, which holds without exception across the particle table. In the edge formulation the term reads endpoint values rather than differences, and the resulting loss of shift invariance is measured.

Not established. Any particular mass value, or any relation among them. The attempt to obtain the ratios from an operator spectrum failed against every candidate, and the failure is

informative rather than neutral. The Standard Model has the same gap and expresses it as free Yukawa couplings; this framework does not close it.

The reframed question. The open problem is no longer why mass exists. It is what fixes the coefficient — equivalently, what fixes the rate of the internal clock. The form of the equation is constrained by the continuation requirement. The spectrum is not, and nothing in this paper suggests it soon will be.

11 Open Edges

What selects the coefficient. Stated above as the central remaining problem. Any candidate must reproduce ratios that no simple spectrum produces, which is a strong constraint on proposals.

Whether the boundedness clause of Section 2 can be derived rather than added. The requirement that a persistent state admit a bounded solution is physically motivated — unbounded configurations carry unbounded energy — but it is imposed here rather than obtained from the continuation constraint alone.

Whether the argument extends beyond scalar fields. Everything here concerns a scalar field on a fixed background. Fermion masses arise through a different mechanism, and gauge boson masses through another again. Whether the same necessity argument applies to those cases is untested.

Whether the massless runaway branch is physically meaningful. The linear-in-time solution of the uniform massless mode is excluded here as non-persistent. In gauge theories such zero modes are typically removed by gauge fixing rather than by dynamics, and whether the exclusion used here is the same exclusion is not settled.

The Koide relation. Unexplained, precise, sector-specific, and not addressed by anything in this paper.

Status Ledger

Claim	Status	Basis (live measurement)
Uniform massless mode is static or unbounded	PROVEN	1.0000 held; 1.0010 → 5.0000 climbing

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Claim	Status	Basis (live measurement)
Uniform massive mode is bounded, both branches	PROVEN	$\max \phi = 1.0002$ and 1.0000
Mass term makes the gradient-free mode persist	PROVEN	Only case of four that persists and evolves
Oscillation rate equals the mass coefficient	PROVEN	predicted 20.0000, measured 20.0011
Massless $\Leftrightarrow$ no rest frame	PROVEN	9 of 9 particles, no exceptions
Mass term breaks shift symmetry	PROVEN	0.2000 invariant vs $-0.8 \rightarrow -4.8$
Uniform massless field is frozen	RETRACTED	Imprecise; static OR runaway. Section 4
Mass ratios from a natural operator spectrum	FAILED	206.8 observed vs 5.27 best candidate
Koide as evidence for the framework	NULL	$1.2e-05$ of random triples match as well
Koide as a predictive constraint	MEASURED	tau to 0.038% from the other two
What selects the coefficient	OPEN	Central remaining problem
Boundedness clause derived, not imposed	OPEN	Physically motivated, not obtained
Extension beyond scalar fields	OPEN	Fermion, gauge mechanisms untested

Appendix: Live Output, Quoted Verbatim

Section 4 — the four uniform configurations.

case	phi start	phi end	phi max
massless, B=0 (static)	1.0000	1.0000	1.0000
massless, B!=0 (drift)	1.0010	5.0000	5.0000
massive, B=0	0.9984	-0.9821	1.0002
massive, B!=0	0.9994	-0.9769	1.0000

Section 5 — the rate.

simulated $m^2 = 400.0$, predicted $\omega = 20.0000$
 measured ω from zero crossings = 20.0011 error 0.006%
 electron: $7.7634e+20$ rad/s period $8.0933e-21$ s
 proton: $1.4255e+24$ rad/s period $4.4077e-24$ s

Section 7 — the shift test.

K added	massless J	massive J
0.0	0.2000	-0.8000
1.0	0.2000	-1.2000
5.0	0.2000	-2.8000
10.0	0.2000	-4.8000

Section 8 — the failed spectrum search.

operator	r21	r32	r31
OBSERVED (leptons)	206.768	16.817	3477.228
particle in a box (n^2)	4.000	2.250	9.000
harmonic oscillator ($n+1/2$)	3.000	1.667	5.000
sphere Laplacian $l(l+1)$	3.000	2.000	6.000
circular drum (Bessel J_0)	5.269	2.458	12.949

log spacings: 5.3316, 2.8224 -> differ by 47%, NOT geometric

Section 9 — Koide with controls.

measured $Q = 0.6666605115$ $2/3 = 0.6666666667$ relative dev $9.233e-06$
 up-type quarks $Q = 0.848981$ down-type quarks $Q = 0.731428$
 random triples closer to $2/3$ than leptons: $24/2000000 = 1.200e-05$

The incomplete Universe

fraction of random triples within 0.01 of 2/3: 0.0234

fit theta to $m_\mu/m_e \rightarrow$ PREDICTED $m_\tau = 1777.535$ MeV

OBSERVED $m_\tau = 1776.860$ MeV error = 0.0380 %

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